

KELT-23Ab

R-2024

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-23A_260504 · created 2026-08-08T00:22:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461164.726 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.024 +/- 0.048
Transit depth (Rp/Rs)^2	0.056 +/- 0.23 [%]
Orbital Inclination (inc)	82.2 +/- 2.58
Airmass coefficient 1 (a1)	0.24 +/- 0.14
Airmass coefficient 2 (a2)	1.5 +/- 0.56
Scatter in the residuals of the lightcurve fit is	49.91 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	3.13
Transit Duration (day)	0.03 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.024 ± 0.048 vs 0.132 (deviation 1.16σ)
Duration fit vs published	0.03 d vs 0.0954 d (deviation 1.0σ)
Mid-time O–C	-2.4 min
Depth SNR	0.3
Residual scatter	49.91 %

Light curve

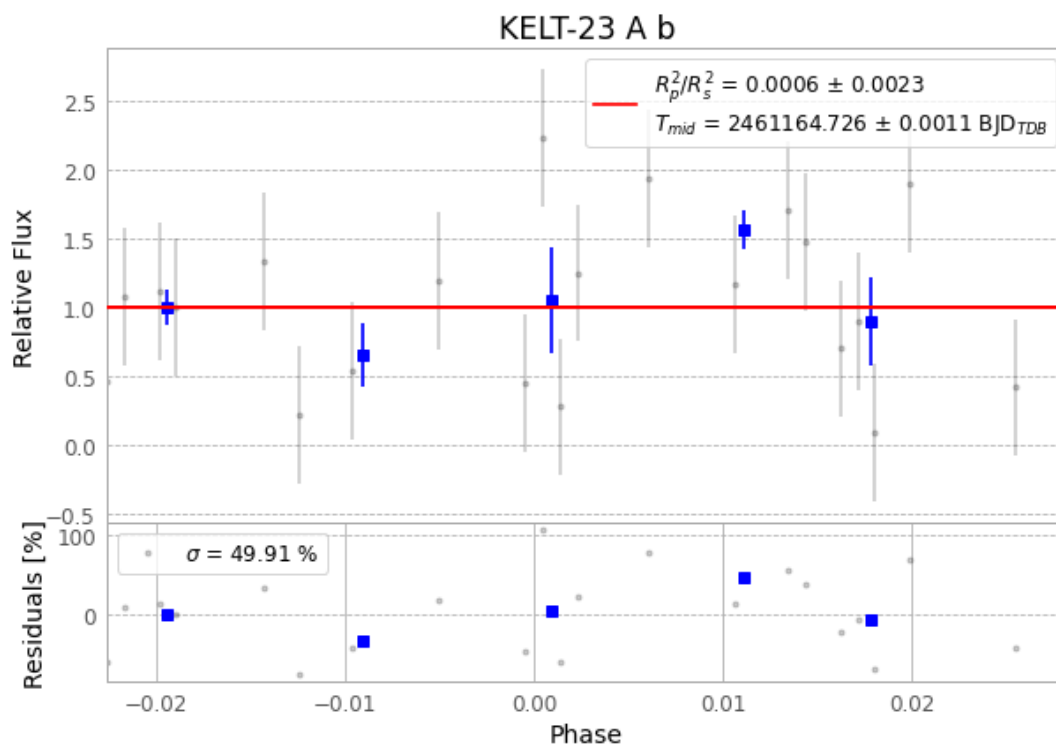


Figure 1 — FinalLightCurve_KELT-23 A b_2026-05-03.png (SHA-256 91682b77ccc47a9a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [275, 84], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f44e75d2801a2741...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

56 FITS frames (37 MB), KELT-23A260504040916.FITS → KELT-23A260504065415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-23A-260504.log (9b17bcba94b1...), FinalLightCurve_KELT-23 A b_2026-05-03.png (91682b77ccc4...), FinalLightCurve_KELT-23 A b_2026-05-03.csv (8e9e5271af83...)

Compute resources

Wall time 136.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 00:22Z.